Euclid's Elements — Book II
Proposition 8

Formal Reconstruction — Technical Documentation

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If a straight line is cut at an arbitrary point, four times the rectangle contained by the whole and one of the parts, together with the square on the remaining part, equals the square described on the whole and the chosen part taken together as one straight line.

1 Introduction

Proposition II.8 is the first proposition in the current Book II reconstruction whose classical statement explicitly asks for four copies of one rectangle. Let C lie between A and B. Euclid extends the line beyond B to a point D such that $BD = BC$. The proposition then asserts

$$4\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

Here the expression "four times" is geometric language for four equal plane figures. In the Lean theorem it becomes four literal copies of one concrete scissors term; there is no scalar multiplication operation on content terms.

A modern algebraic shadow is obtained by writing $AC = a$ and $CB = b$. Then $AB = a + b$ and the produced condition $BD = BC$ gives $AD = a + 2b$, so the mnemonic is

$$4(a + b)b + a^2 = (a + 2b)^2.$$

This identity is documentary only. Neither the classical proof nor the current formal proof introduces numerical area or multiplication of segment lengths.

The production source begins with

```
import CGJteamLab.Proposition2_7
```

so Proposition II.7 is the immediate module dependency. At theorem-call level the proof uses `euclid_proposition_2_4` and `euclid_proposition_2_7`, together with the established rectangle representative API and exact scissors addition, symmetry, and transitivity.

2 Euclid's Statement and Classical Configuration

2.1 The statement

Euclid's statement says that if a straight line is cut at random, four times the rectangle contained by the whole and one segment, together with the square on the remaining segment, is equal to the square described on the whole and the chosen segment taken together as one straight line [**?**].

With the current lettering, $A - C - B$ and the chosen part is BC. Euclid produces D beyond B with $BD = BC$. Thus the final square is the square on AD, and the geometric identity is

$$4\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

The phrase "on the whole and the chosen part as one straight line" is therefore not an abstract addition of lengths: it is represented by the actually produced collinear segment AD.

2.2 The classical diagram

Euclid produces BD in a straight line with AB, makes $BD = CB$, describes the square on AD, and draws the network of parallels usually called the "double" figure. The proof identifies two families of four equal regions by I.34, I.36, and I.43. Together the eight regions form a gnomon equal to four copies of the rectangle contained by AB and BD. Adding the remaining square on AC fills the square on AD [**?**].

Figure 1 deliberately separates that classical produced segment from the normal forms used by Lean. The middle panel is the II.4 normal form on $A - B - D$; the right panel records the final formal sum rather than pretending that the production proof replays Euclid's eight-region gnomon.

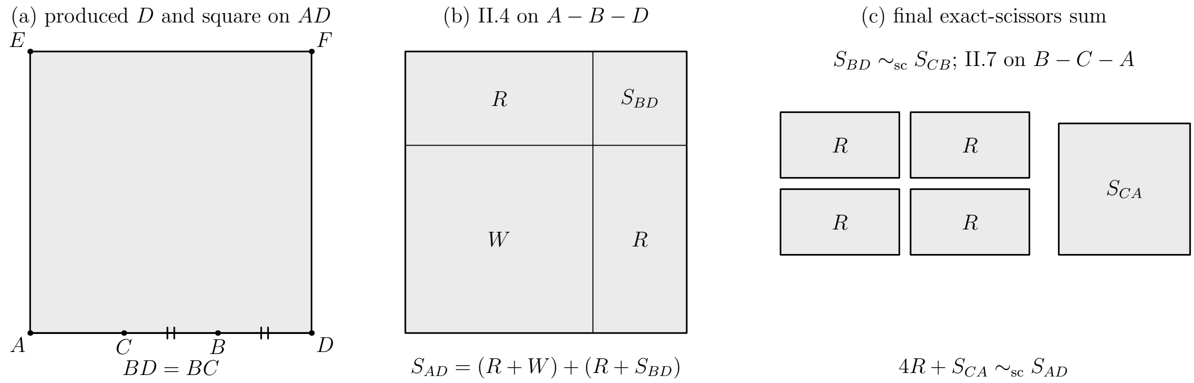

Figure 1: The mechanism used to read Proposition II.8. Left: the baseline order $A - C - B - D$ with $BD = BC$ and the square on AD. Middle: II.4 on $A - B - D$ gives $\mathrm{Sq}(AD) = (R + \mathrm{Sq}(BA)) + (R + \mathrm{Sq}(BD))$. Right: after transporting the square on BD to the square on CB and applying II.7 to $B - C - A$, the formal sum becomes four copies of the same rectangle term R plus the square on CA. Region labels denote concrete scissors terms, not numerical areas.

3 Classical Proof

Start with $A - C - B$ and produce D beyond B so that $BD = BC$. Construct the square $AEFD$ on AD and draw Euclid's double network of parallels. The proof then has two four-region stages [**?**].

First, the equalities $CB = BD$ are propagated through opposite sides of the constructed parallelograms. I.36 converts the equal bases into equal parallelograms, while I.43 identifies the relevant complements. Euclid obtains four equal regions DK, CK, GR, RN.

A second use of the same mechanism gives four equal regions AG, MQ, QL, RF. The eight regions together form the gnomon STU, and Euclid proves that this gnomon is four times the region AK. The latter is a rectangle contained by AB and BD, because one of its adjacent sides is equal to BD. Consequently

$$4\ \mathrm{Rect}(AB, BD) = \text{gnomon } STU.$$

The remaining region OH is the square on AC. Adding it to both sides gives

$$4\ \mathrm{Rect}(AB, BD) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

Finally $BD = BC$, so the rectangle may be read as the rectangle contained by AB and BC.

The classical local dependency picture is therefore

$$\begin{array}{c}
\text{I.3: lay off } BD = BC \quad + \quad \text{I.46: square on } AD \\
\Downarrow \\
\text{I.31: the parallel network} \\
\Downarrow \\
\text{I.34 and I.36: side and parallelogram equalities} \\
\Downarrow \\
\text{I.43: equal complements} \\
\Downarrow \\
\text{eight-region gnomon and common-figure addition} \\
\Downarrow \\
\text{II.8.}
\end{array}$$

The Lean DAG below is substantially shorter because the relevant Book II normal forms have already been proved.

4 The Geometric Identity in the Current Formalization

The public theorem receives the three baseline facts

```
(A B C D : Geo.Point)
(hACB : Geo.Between A C B)
(hABD : Geo.Between A B D)
(hBD_BC : Geo.Congruent B D B C)
```

so the intended order is

$$A - C - B - D, \qquad BD \cong BC.$$

Only the two displayed betweenness relations are primitive inputs; the proof reverses $A - C - B$ explicitly when II.7 needs $B - C - A$.

Let

$$R := \mathrm{Rect}(AB, BD).$$

The same concrete quadrilateral is then re-read as $\mathrm{Rect}(BA, BC)$ using endpoint reversal on the first containing side and the hypothesis $BD \cong BC$ on the second. The public conclusion is exactly

$$(R + R) + (R + R) + \mathrm{Sq}(CA) \simeq_{\mathrm{sc}} \mathrm{Sq}(AD).$$

There is no hidden coefficient object: four occurrences of R appear literally in the Lean term.

This conclusion is `HilbertScissorsEq`, not merely `HilbertScissorsEquicomplementable`. The theorem therefore certifies an exact finite-scissors relation for the configured representatives. It does not assert equality of real numbers.

5 Reusable Book II Infrastructure Used

5.1 Proposition II.4 on the produced segment

The first normal form is obtained by applying II.4 to the whole segment AD cut at B:

$$\mathrm{Sq}(AD) \simeq_{\mathrm{sc}} (R + \mathrm{Sq}(BA)) + (R + \mathrm{Sq}(BD)).$$

In this invocation the same concrete representative $T_0T_1T_2T_3$ is supplied to both cross-rectangle positions. Consequently the two occurrences of R are literally identical scissors terms.

5.2 Proposition II.7 on the reversed cut

From $A - C - B$ the proof derives $B - C - A$ and applies II.7 with the whole segment read as BA. The result is

$$\mathrm{Sq}(BA) + \mathrm{Sq}(CB) \simeq_{\mathrm{sc}} (R + R) + \mathrm{Sq}(CA).$$

The square on BA is deliberately the same concrete square $BAFG$ already used inside the II.4 call. This avoids a second representative-transport step.

5.3 Square transport through the rectangle API

II.4 supplies a square on BD, whereas II.7 uses a square on CB. The proof converts both squares to `IsRectangleContainedBy` data with `square_is_rectangle_contained_by_side`. The congruence $BD \cong BC$ is then used to read the CB square as another representative of the same rectangle contained by BD, BD. Finally `rectangle_contained_by_unique` gives

$$\mathrm{Sq}(BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

This is representative transport in the established geometric rectangle API; it is not an appeal to numerical equality of squared lengths.

5.4 Exact scissors addition and regrouping

After the three normal forms are available, the remainder is formal addition, transitivity, symmetry, and associativity/commutativity of the multiset sum. No cancellation, subtraction, positivity theorem, or numerical content map is used.

6 Proof Architecture of the Reconstruction

The production proof can be read in seven steps.

6.1 Step 1: reverse the internal cut

The source assumption $A - C - B$ is converted to $B - C - A$ by

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the orientation required by the II.7 invocation.

6.2 Step 2: re-read the repeated rectangle

The concrete rectangle $T_0T_1T_2T_3$ initially satisfies

$$\mathrm{Rect}(AB, BD).$$

The proof reverses AB to BA with `CongruentSwapSecond` and transports BD to BC by congruence transitivity. The same quadrilateral therefore satisfies

$$\mathrm{Rect}(BA, BC).$$

No scissors theorem is needed because the geometric representative itself is unchanged.

6.3 Step 3: apply II.4 to $A - B - D$

The theorem `euclid_proposition_2_4` produces

$$\mathrm{Sq}(AD) \simeq_{\mathrm{sc}} (R + W) + (R + S_{BD}),$$

where

$$W := \mathrm{Sq}(BA), \qquad S_{BD} := \mathrm{Sq}(BD).$$

6.4 Step 4: apply II.7 to $B - C - A$

The theorem `euclid_proposition_2_7` produces

$$W + S_{CB} \simeq_{\mathrm{sc}} (R + R) + S_{CA},$$

where

$$S_{CB} := \mathrm{Sq}(CB), \qquad S_{CA} := \mathrm{Sq}(CA).$$

6.5 Step 5: transport the square on the congruent produced segment

Using the square-as-rectangle conversion and `rectangle_contained_by_unique`, the proof obtains

$$S_{BD} \simeq_{\mathrm{sc}} S_{CB}.$$

This is the formal counterpart of Euclid's final use of $BD = BC$.

6.6 Step 6: substitute and regroup

Scissors addition first replaces S_{BD} by S_{CB} inside the II.4 normal form. Associativity and commutativity then expose the summand $W + S_{CB}$:

$$(R + W) + (R + S_{CB}) = (R + R) + (W + S_{CB}).$$

The proof uses `ac_rfl`; this is equality of the formal multiset sum, not a new geometric theorem.

6.7 Step 7: replace the II.7 block and close

Replacing $W + S_{CB}$ by the II.7 normal form gives

$$\begin{aligned}\mathrm{Sq}(AD) &\simeq_{\mathrm{sc}} (R + R) + ((R + R) + S_{CA}) \\ &= ((R + R) + (R + R)) + S_{CA}.\end{aligned}$$

A final symmetry puts the four rectangles plus the square on the left, exactly in the orientation of the public theorem.

7 Lean Formalization

7.1 The public theorem signature

The current compiling source declares:

```
theorem euclid_proposition_2_8
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)

    (hACB : Geo.Between A C B)
    (hABD : Geo.Between A B D)
    (hBD_BC : Geo.Congruent B D B C)

    -- ==================================================================
    -- II.4 on A-B-D.
    -- ==================================================================

    (D40 E40 L40 X40 : Geo.Point)
    (hSquareAD : IsSquare Geo A D E40 D40)

    (hD40L40E40 : Geo.Between D40 L40 E40)
    (hD40X40D : Geo.Between D40 X40 D)
    (hBX40L40 : Geo.Between B X40 L40)

    (hII4LeftPar :
      IsParallelogram Geo L40 D40 A B)

    (hII4RightPar :
      IsParallelogram Geo D E40 L40 B)

    -- Rect(AD,AB), Rect(AD,BD).
```

```
(U40 U41 U42 U43 : Geo.Point)
(V40 V41 V42 V43 : Geo.Point)

(hAD_AB :
  IsRectangleContainedBy Geo
    U40 U41 U42 U43 A D A B)

(hAD_BD :
  IsRectangleContainedBy Geo
    V40 V41 V42 V43 A D B D)

-------------------------------------------------------------
-- Left II.3 branch inside II.4.
-------------------------------------------------------------

(D41 E41 L41 X41 : Geo.Point)

(hRect41 :
  IsRectangle Geo D A E41 D41)

(hAE41_BA :
  Geo.Congruent A E41 B A)

(hD41L41E41 : Geo.Between D41 L41 E41)
(hD41X41A : Geo.Between D41 X41 A)
(hBX41L41 : Geo.Between B X41 L41)

(hLeft41 :
  IsParallelogram Geo L41 D41 D B)

(hRight41 :
  IsParallelogram Geo A E41 L41 B)

-------------------------------------------------------------
-- Common square on BA: used here and again as the outer square
-- in the II.7 invocation below.
-------------------------------------------------------------

(F G : Geo.Point)
(hSquareBA : IsSquare Geo B A F G)

-------------------------------------------------------------
-- Right II.3 branch inside II.4.
-------------------------------------------------------------

(D42 E42 L42 X42 : Geo.Point)

(hRect42 :
  IsRectangle Geo A D E42 D42)

(hDE42_BD :
  Geo.Congruent D E42 B D)

(hD42L42E42 : Geo.Between D42 L42 E42)
(hD42X42D : Geo.Between D42 X42 D)
(hBX42L42 : Geo.Between B X42 L42)

(hLeft42 :
```

```
  IsParallelogram Geo L42 D42 A B)

(hRight42 :
  IsParallelogram Geo D E42 L42 B)

(H K : Geo.Point)
(hSquareBD : IsSquare Geo B D H K)

-------------------------------------------------------------
-- Common repeated rectangle:
--
--   in II.4: Rect(AB,BD),
--   in II.7: Rect(BA,BC).
-------------------------------------------------------------

(T0 T1 T2 T3 : Geo.Point)

(hCross4 :
  IsRectangleContainedBy Geo
    T0 T1 T2 T3 A B B D)

-- =============================================================
-- II.7 on the reversed division B-C-A.
--
-- The outer square is exactly the common square B-A-F-G above.
-- =============================================================

(L70 X70 : Geo.Point)

(hGL70F : Geo.Between G L70 F)
(hGX70A : Geo.Between G X70 A)
(hCX70L70 : Geo.Between C X70 L70)

(hII7LeftPar :
  IsParallelogram Geo L70 G B C)

(hII7RightPar :
  IsParallelogram Geo A F L70 C)

-- Second outer representative Rect(BA,CA).
(V70 V71 V72 V73 : Geo.Point)

(hBA_CA :
  IsRectangleContainedBy Geo
    V70 V71 V72 V73 B A C A)

-------------------------------------------------------------
-- Left II.3 branch inside II.7.
-------------------------------------------------------------

(D71 E71 L71 X71 : Geo.Point)

(hRect71 :
  IsRectangle Geo A B E71 D71)

(hBE71_CB :
  Geo.Congruent B E71 C B)
```

```
  (hD71L71E71 : Geo.Between D71 L71 E71)
  (hD71X71B : Geo.Between D71 X71 B)
  (hCX71L71 : Geo.Between C X71 L71)

  (hLeft71 :
    IsParallelogram Geo L71 D71 A C)

  (hRight71 :
    IsParallelogram Geo B E71 L71 C)

  (M N : Geo.Point)
  (hSquareCB : IsSquare Geo C B M N)

  --------------------------------------------------------------
  -- Right II.3 branch inside II.7.
  --------------------------------------------------------------

  (D72 E72 L72 X72 : Geo.Point)

  (hRect72 :
    IsRectangle Geo B A E72 D72)

  (hAE72_CA :
    Geo.Congruent A E72 C A)

  (hD72L72E72 : Geo.Between D72 L72 E72)
  (hD72X72A : Geo.Between D72 X72 A)
  (hCX72L72 : Geo.Between C X72 L72)

  (hLeft72 :
    IsParallelogram Geo L72 D72 B C)

  (hRight72 :
    IsParallelogram Geo A E72 L72 C)

  (P Q : Geo.Point)
  (hSquareCA : IsSquare Geo C A P Q)

  --------------------------------------------------------------
  -- Cross rectangle internal to II.7: Rect(BC,CA).
  --------------------------------------------------------------

  (Z0 Z1 Z2 Z3 : Geo.Point)

  (hCross7 :
    IsRectangleContainedBy Geo
      Z0 Z1 Z2 Z3 B C C A) :

  HilbertScissorsEq Geo
    (((hilbertParallelogramTerm Geo T0 T1 T2 T3 +
       hilbertParallelogramTerm Geo T0 T1 T2 T3) +
      (hilbertParallelogramTerm Geo T0 T1 T2 T3 +
       hilbertParallelogramTerm Geo T0 T1 T2 T3)) +
     hilbertParallelogramTerm Geo C A P Q)
    (hilbertParallelogramTerm Geo A D E40 D40)
```

The signature is intentionally large because the theorem is configured around concrete II.4 and

II.7 diagrams. The meaningful top-level geometric data are small: $A - C - B$, $A - B - D$, and $BD \cong BC$. The remaining points certify the squares, rectangles, cuts, and parallelograms consumed by the already proved Book II propositions.

7.2 The decisive local composition

The two principal theorem calls are

```
have hII4 :=
  euclid_proposition_2_4 Geo A D B ...

have hII7 :=
  euclid_proposition_2_7 Geo B A C ...
```

The ellipses here are documentary abbreviations only; the full public signature above records the actual configured witnesses.

The exact square transport is obtained with

```
have hSquareTransport :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B D H K)
      (hilbertParallelogramTerm Geo C B M N) :=
  rectangle_contained_by_unique
    Geo
    B D H K
    C B M N
    B D B D
    hSquareBDContained
    hSquareCBAsBD
```

so no new II.8-specific transport theorem is added to the library.

8 Relationship with Earlier Book II Propositions

8.1 Proposition II.4

II.4 is called directly on the produced whole AD cut at B. It supplies the square decomposition containing two copies of $\mathrm{Rect}(AB, BD)$, the square on BA, and the square on BD. This is the left normal form of the production proof.

8.2 Proposition II.7

II.7 is called directly on the same original arbitrary cut but in reversed orientation $B - C - A$. Its selected part is therefore BC, giving precisely

$$\mathrm{Sq}(BA) + \mathrm{Sq}(CB) \simeq_{\mathrm{sc}} 2\ \mathrm{Rect}(BA, BC) + \mathrm{Sq}(CA).$$

This is the block substituted into the II.4 expansion.

8.3 Propositions II.1–II.3

II.1–II.3 are not called directly by II.8. They lie below II.4 and II.7 in the established rectangle-calculus DAG. In particular, the direct proof does not need to reopen the binary rectangle split or the II.3 square specialization.

8.4 Propositions II.5 and II.6

II.5 and II.6 are not dependencies of II.8. Their midpoint hypotheses belong to a different Book II block. II.8 uses an arbitrary internal cut and a newly produced congruent external segment, not a midpoint.

9 Relationship with Book I Infrastructure

Classically, II.8 explicitly depends on I.3, I.31, I.34, I.36, I.43, and I.46: segment layoff, parallel construction, parallelogram side equalities, equal parallelograms on equal bases, equal complements, and square construction [?]. The present production theorem does not call those propositions locally. Their consequences have already been packaged in the square, rectangle, parallelogram, congruence, and exact-scissors infrastructure consumed by II.4 and II.7.

This is a substantial classical-DAG/Lean-DAG difference. The formal proof is not a transcription of the eight-region gnomon argument. It is a composition of previously verified geometric identities.

10 Formalization Notes and Hidden Geometric Data

10.1 The line order is split across two hypotheses

The public theorem does not take a single four-point order predicate. It takes $A - C - B$ and $A - B - D$. Together these encode the intended baseline $A - C - B - D$. The proof only derives the reversed $B - C - A$ fact it actually needs.

10.2 The same rectangle has two contained-by descriptions

The term $T_0T_1T_2T_3$ first represents $\mathrm{Rect}(AB, BD)$ and later represents $\mathrm{Rect}(BA, BC)$. The first change is endpoint orientation; the second is metric transport through $BD \cong BC$. This is representation data, not commutativity of a primitive segment product.

10.3 Representative reuse is deliberate

The same square on BA is supplied to both II.4 and II.7, and the same rectangle term R is supplied repeatedly. This design removes unnecessary applications of representative uniqueness from the final scissors bridge. Only the genuinely different squares on BD and CB require transport.

10.4 Four times means four summands

The public conclusion contains four literal occurrences of `hilbertParallelogramTerm` for the same rectangle. No natural-number scalar action is introduced. The proof therefore stays in the finite formal sum language already used throughout the scissors layer.

10.5 No cancellation or positivity

Although II.8 looks algebraically like an identity that could be proved by expanding and cancelling terms, the Lean proof performs no cancellation. It uses only forward exact-scissors equalities, addition, transitivity, symmetry, and regrouping. No positivity theorem enters.

11 Axiomatic and Semantic Status

The configured public theorem proves exact `HilbertScissorsEq`. It introduces no numerical area function and no multiplication of segment lengths. The formal "algebra" is therefore a calculus of geometric representatives and finite scissors sums.

The explicit axiom audit was

```
'Geometry.euclid_proposition_2_8' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same transitive profile observed for Proposition II.7. In particular, II.8 introduces no new proposition-specific geometric axiom, no new content axiom, and no `sorryAx` dependency.

The semantic layers should therefore be kept distinct:

$$\begin{array}{c}
\text{incidence/order/congruence and Euclidean rectangle geometry} \\
\Downarrow \\
\text{concrete square and rectangle representatives} \\
\Downarrow \\
\text{exact finite-scissors relation} \\
\not\Downarrow \\
\text{numerical area or segment multiplication in this proof.}
\end{array}$$

12 Dependency Summary

The actual production chain is

$$\begin{gathered}\text{Hilbert incidence, order, congruence, Euclidean parallel layer}\\ \Downarrow\\ \text{Book I parallelogram/square/scissors infrastructure}\\ \Downarrow\\ \text{HilbertRectangle representative API}\\ \Downarrow\\ \text{II.1–II.3 rectangle calculus}\\ \Downarrow\\ \text{II.4 and II.7}\\ \Downarrow\\ \text{reorientation } \operatorname{Rect}(AB, BD) \to \operatorname{Rect}(BA, BC)\\ \Downarrow\\ \text{square transport } \operatorname{Sq}(BD) \simeq_{\mathrm{sc}} \operatorname{Sq}(CB)\\ \Downarrow\\ \text{exact scissors addition and regrouping}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_8}.\end{gathered}$$

The implementation checkpoint is:

- new geometric axiom: no;
- new content axiom: no;
- new proposition-local theorem declaration: no;
- new reusable helper theorem: no;
- new `HilbertRectangle` theorem: no;
- new Interface / Book Zero / Scissors theorem: no;
- uses numerical area: no;
- uses segment multiplication: no;
- public theorem compiles: yes;
- `lake build CGJteamLab.Proposition2_8`: clean;
- full `lake build`: clean;
- `sorry`/`admit`: no.

13 Conclusion

Proposition II.8 is a useful demonstration of the point at which the established Book II API begins to replace long classical gnomon bookkeeping. Euclid's proof constructs an eight-region configuration and proves equalities among its pieces through I.34, I.36, and I.43. The Lean proof instead takes two already verified normal forms: II.4 on the produced segment AD and II.7 on the reversed original cut $B - C - A$.

The only genuinely new local bridge is representational. The segment equality $BD \cong BC$ lets the repeated rectangle and the square on the produced segment be read in the forms required by II.7. Once those transports are explicit, the fourfold rectangle identity follows from exact scissors addition and regrouping.

Thus the algebraic-looking statement

$$4(a + b)b + a^2 = (a + 2b)^2$$

is not the proof object. It is a modern mnemonic for a theorem whose formal content remains entirely synthetic: betweenness, congruence, squares, rectangles contained by segments, and exact finite scissors equality.